DT081A – Signal and Image Processing

Laboratory 1 Report

Sections 1.1–1.3, 2.1–2.3 and 3.1 – Signal and Image Processing

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1 Introduction

This part of Lab 1 compares sine, square and sawtooth waves. It shows how frequency and addition change a signal. FFTs were also used to study fixed frequencies, noise and a chirp. I also tested how different sampling rates affect a signal. I also compared image resampling and requantization. I also compared image quality using PSNR and SSIM. I also measured how Gaussian noise and filtering changed image PSNR. I also analyzed audio recordings and tested downsampling.

2 Experimental Setup

NumPy and Matplotlib were used. All signals use the teacher's time vector, `t = np.linspace(0, 1, 1000)`: 1,000 points from 0 to 1 second, including both ends. The starting signals are a 10 Hz sine and a 5 Hz square. Their amplitudes were kept unchanged. For Section 1.2, the original 10 Hz and 20 Hz sines have amplitudes 1 and 0.5. The FFT uses the time step between the given points. I used the teacher's sampling examples and tested a 23 Hz sine wave sampled at 8 Hz. Pillow loaded Lenna and a grass texture as grayscale images. PSNR used floating-point pixel errors, and SSIM used a data range of 255. Gaussian noise used seed 31415, and Gaussian filtering used spatial $\sigma = 1$. SciPy loaded the piano and thunderstorm WAV files; stereo channels were averaged before peak normalization. Spectrograms used 1,024-sample windows with 512-sample overlap; the frequency estimate used 1% of the strongest non-DC FFT magnitude.

3 Results and Analysis

3.1 Signal Generation and Visualization

Q1. Modify the code to create a sawtooth wave. Describe the mathematical function you used.

Answer: The function is $s(t) = 2(5t - \lfloor 5t \rfloor) - 1$. Here, $\lfloor 5t \rfloor$ means rounding down to a whole number. Subtracting it leaves the part of a cycle between 0 and 1. Scaling by 2 and subtracting 1 gives a rising ramp from -1 towards $+1$. It jumps back every 0.2 seconds, so its frequency is 5 Hz (Figure 1).

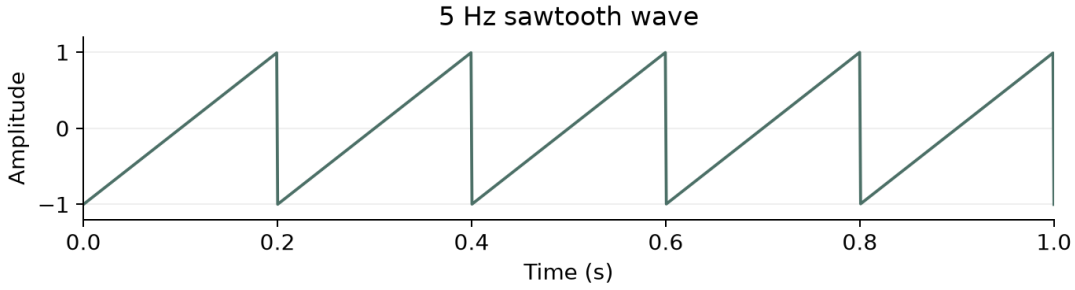


Figure 1: Five rising ramps during one second. Each ramp ends with a jump back.

Q2. Change the frequency of the sine wave to 20 Hz. How does this affect the plot?

Answer: The 20 Hz sine has 20 cycles in one second instead of 10. The waves are closer together in time. The amplitude stays the same (Figure 2).

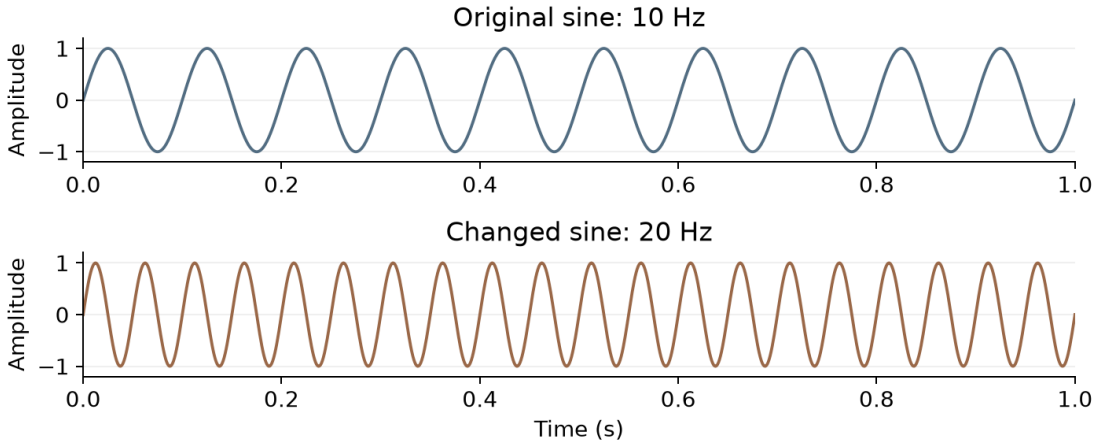


Figure 2: The same one-second interval and amplitude limits show twice as many cycles at 20 Hz.

Q3. What happens to the square wave if you change the sign function to `np.ceil(np.sin(2 * np.pi * 5 * t)) * 2 - 1`? Explain why.

Answer: The two expressions give the same value at 999 of the 1,000 samples. Only the first sample differs: at $t = 0$, `np.sign()` gives 0, while the replacement gives -1.

For a positive sine value, rounding up gives 1, so the replacement gives $+1$. For a negative value strictly between -1 and 0, rounding up gives 0, so the result is -1. At zero, it also gives -1, while `np.sign(0)` gives 0. If the sine is exactly -1, the replacement gives $2(-1) - 1 = -3$. No sampled sine value was exactly -1, so -3 did not appear in these results.

Q4. Create a composite signal by adding the sine wave and square wave together. Plot this new signal and describe its characteristics.

Answer: The original 10 Hz sine was added to the 5 Hz square. The sine gives the curved parts, and the square gives sudden jumps. The sampled values range from about -2 to $+2$ (Figure 3).

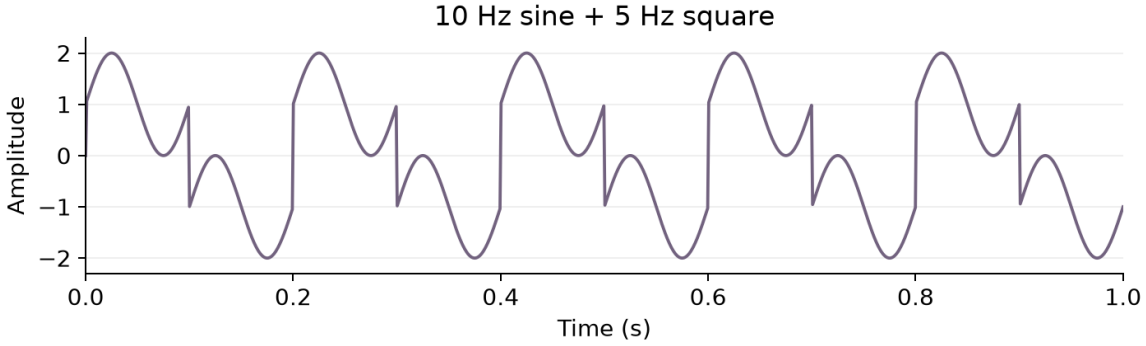


Figure 3: The original sine and square added together, without changing their amplitudes.

Q5. How might understanding different signal types be useful in real-world applications? Provide an example.

Answer: A square wave can act as a clock signal in a digital circuit. Its regular changes tell parts of the circuit when to act. Its frequency tells us how often these changes repeat.

3.2 Fourier Transform and Frequency Domain Analysis

Q1. What frequencies are present in the signal? How can you identify them from the frequency domain plot?

Answer: The signal contains 10 Hz and 20 Hz. They give the two largest FFT peaks, at 9.99 and 19.98 Hz (Figure 4). Using the actual time step gives FFT bins 0.999 Hz apart. The peaks are at the nearest bins to the input frequencies.

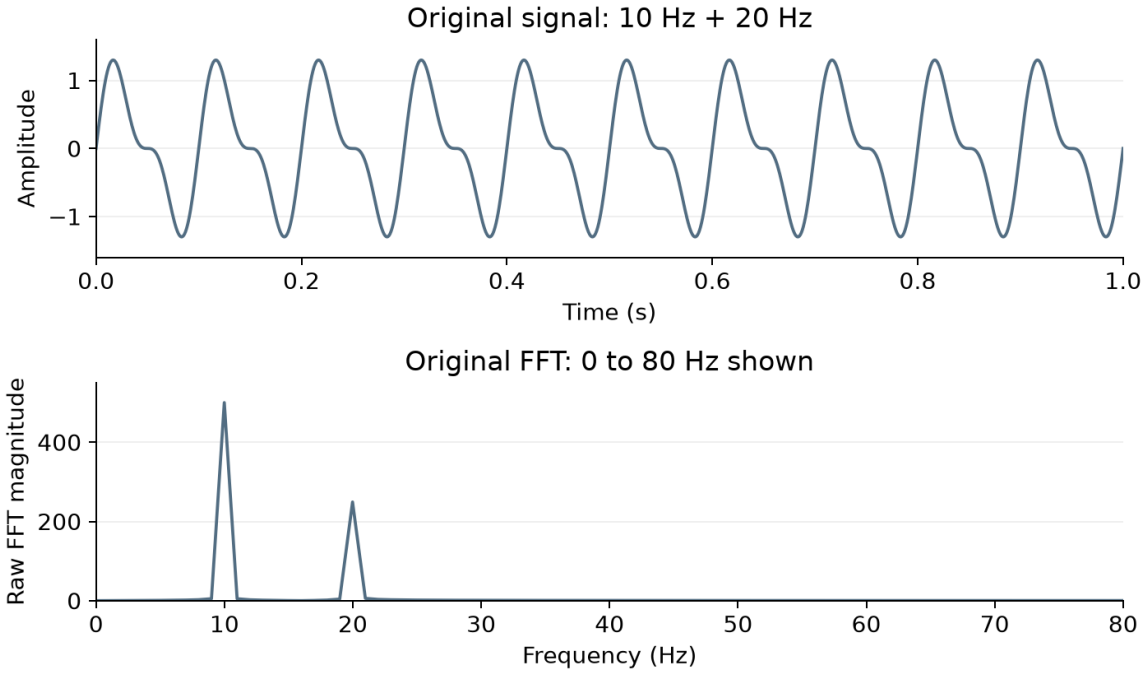


Figure 4: The original two-sine signal and its FFT magnitude. The frequency plot shows 0–80 Hz.

Q2. Modify the generate_signal function to include a 30 Hz component. How does this change the frequency domain representation?

Answer: A 30 Hz sine with amplitude 0.3 was added. A new peak appears at 29.97 Hz, with a raw FFT magnitude of 148.72. The 10 Hz and 20 Hz peaks remain (Figure 5).

Q3. What happens to the frequency domain plot if you add random noise to the signal? Implement this and explain your observations.

Answer: Gaussian noise with standard deviation 0.18 and seed 1729 was added to the original signal. Small, irregular values appear across the spectrum, while the two largest peaks stay at 9.99 and 19.98 Hz. This happens because the noise contains many frequencies (Figure 5).

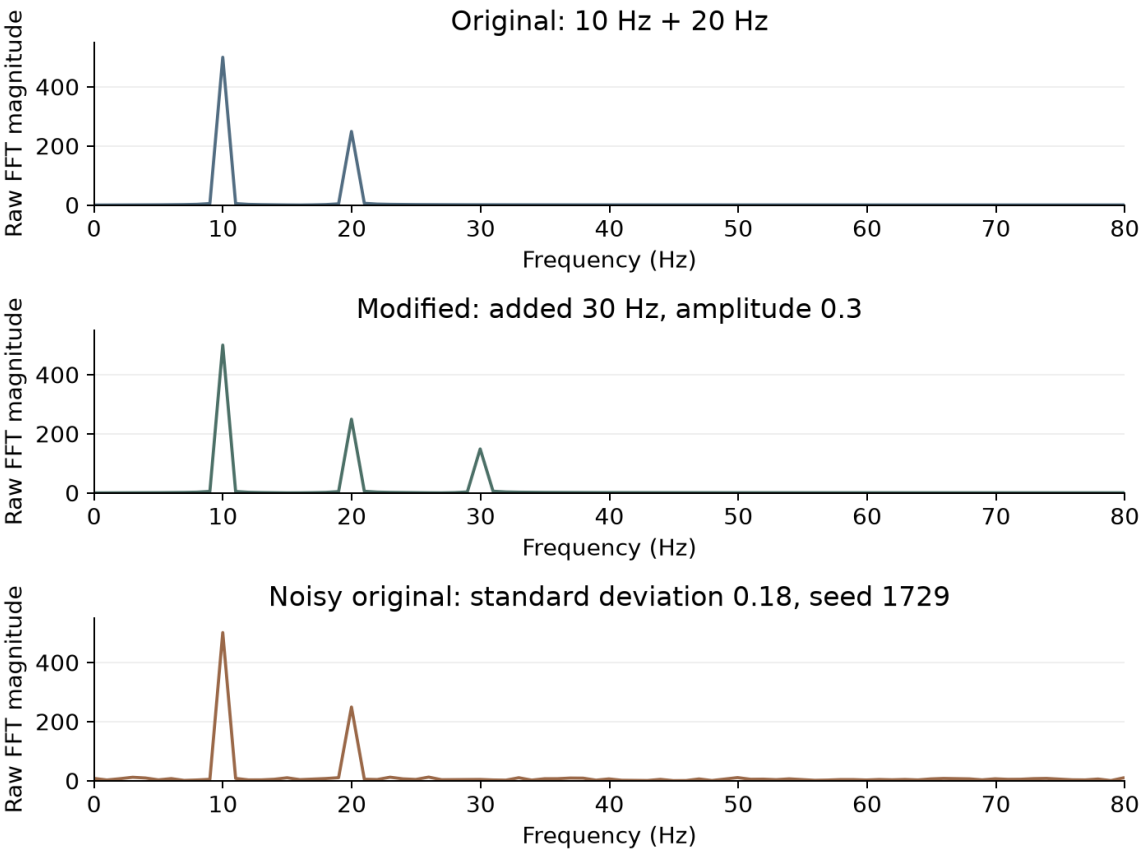


Figure 5: Original, modified and noisy spectra, using the same frequency and magnitude limits. Noise was added to the original signal.

Q4. Create a function that generates a signal with a frequency that changes over time (e.g., a chirp signal). Plot this signal in both time and frequency domains and explain what you observe.

Answer: The function sweeps from 15 Hz to 65 Hz in one second. The waves get closer together over time. The FFT shows a broad band around this range instead of two narrow peaks. It shows which frequencies are present, but not when each one occurs (Figure 6).

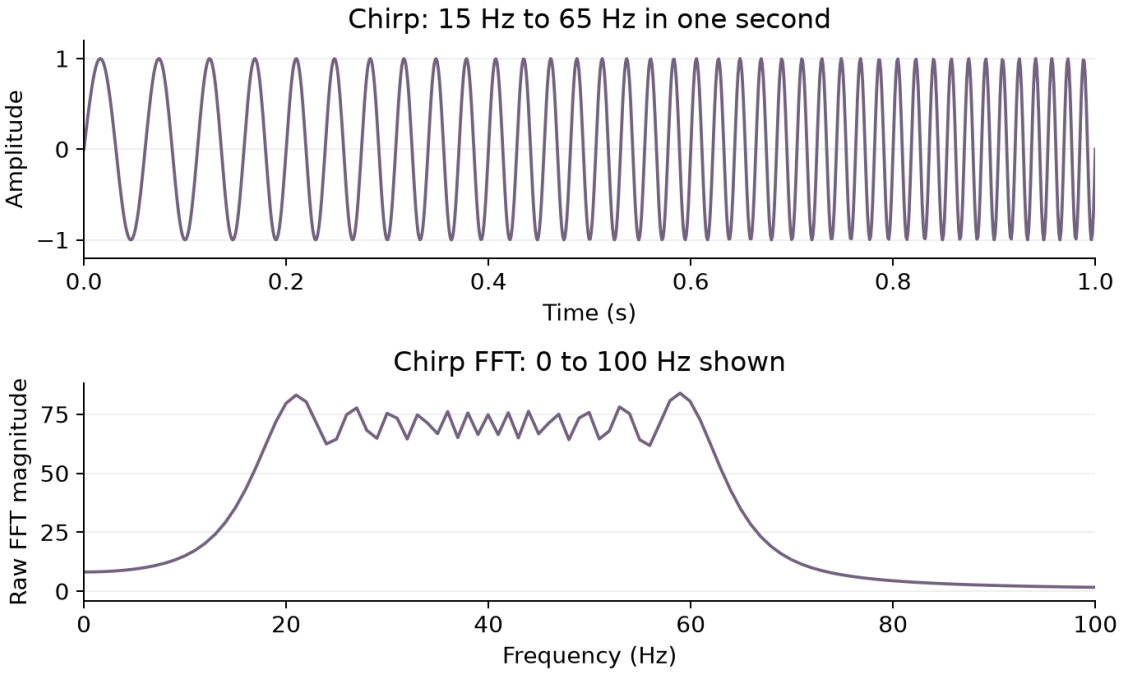


Figure 6: The chirp speeds up during the recording. Its FFT magnitude is shown over 0–100 Hz.

Q5. Describe a practical application where understanding the frequency domain of a signal is important.

Answer: A sensor can measure a machine's vibration. Its frequency peaks can be compared with a normal reading. New or much stronger peaks can help find a fault.

3.3 Sampling and Aliasing

Q1. What is the Nyquist rate for the given signal? Is the current sampling rate sufficient?

Answer: For a 10 Hz wave, the Nyquist rate is 20 Hz. The given 15 Hz sampling rate is too low, so the sampled signal shows a false 5 Hz pattern (Figure 7).

Q2. Increase the sampling rate to 25 Hz. How does this affect the sampled signal's representation of the original signal?

Answer: I increased the sampling rate to 25 Hz. This is above the 20 Hz Nyquist rate, so there is no aliasing. The extra sample points represent the original 10 Hz wave better (Figure 7).

Q3. Set the signal frequency to 20 Hz and the sampling rate to 30 Hz. Describe and explain the resulting aliasing effect.

Answer: I sampled a 20 Hz sine wave at 30 Hz. This is below the required 40 Hz Nyquist rate. The sampled points behave like a -10 Hz sine wave because the sampling rate is too low (Figure 7).

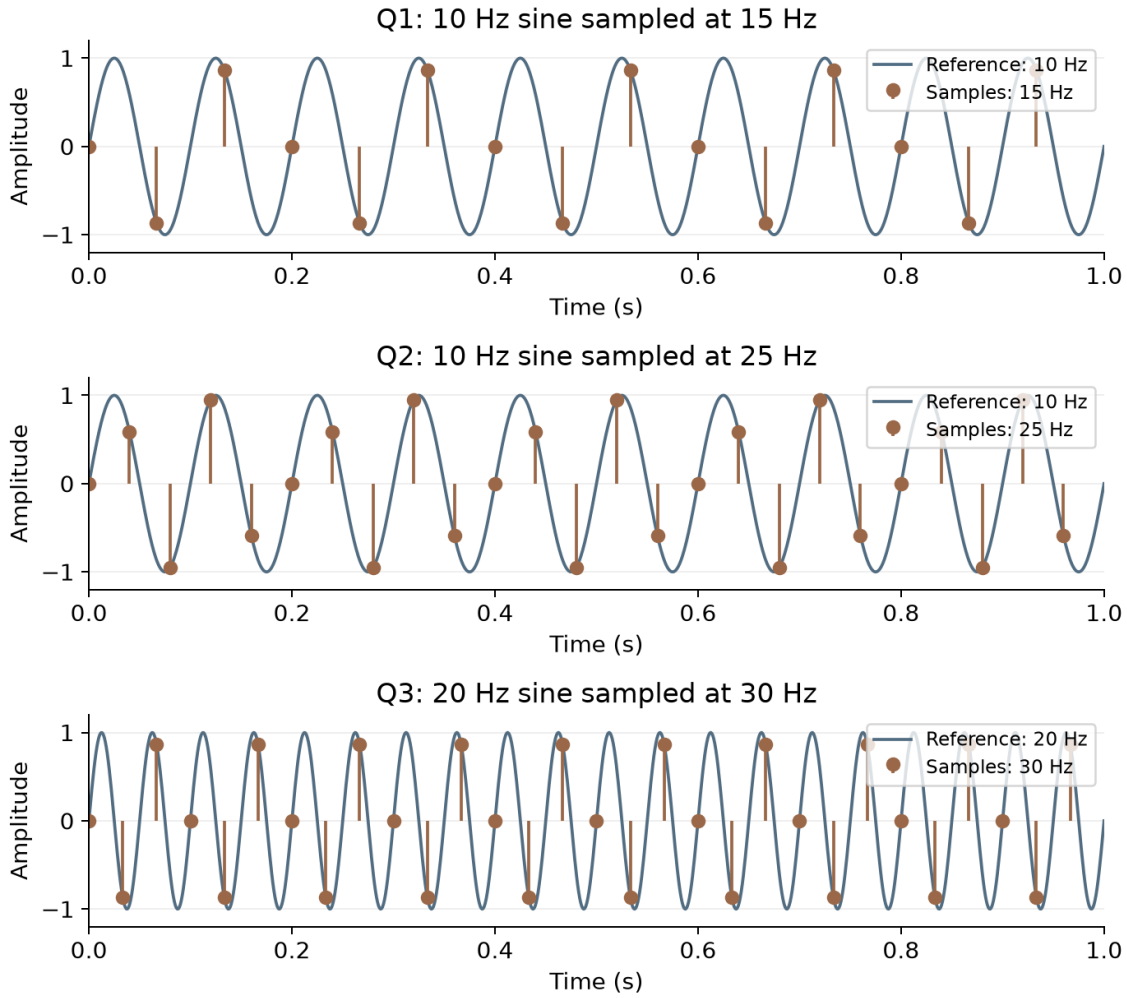


Figure 7: Sampling comparison for the three given cases. The 25 Hz sampling rate represents the 10 Hz signal without aliasing.

Q4. Create a function that demonstrates the "wagon-wheel effect" by sampling a high-frequency sinusoid at a much lower rate. Plot the original and sampled signals, and explain the visual effect.

Answer: I sampled a 23 Hz sine wave at 8 Hz. The sampled points match a 1 Hz wave with its sign reversed. This makes a fast repeating signal look much slower, like the wagon-wheel effect (Figure 8).

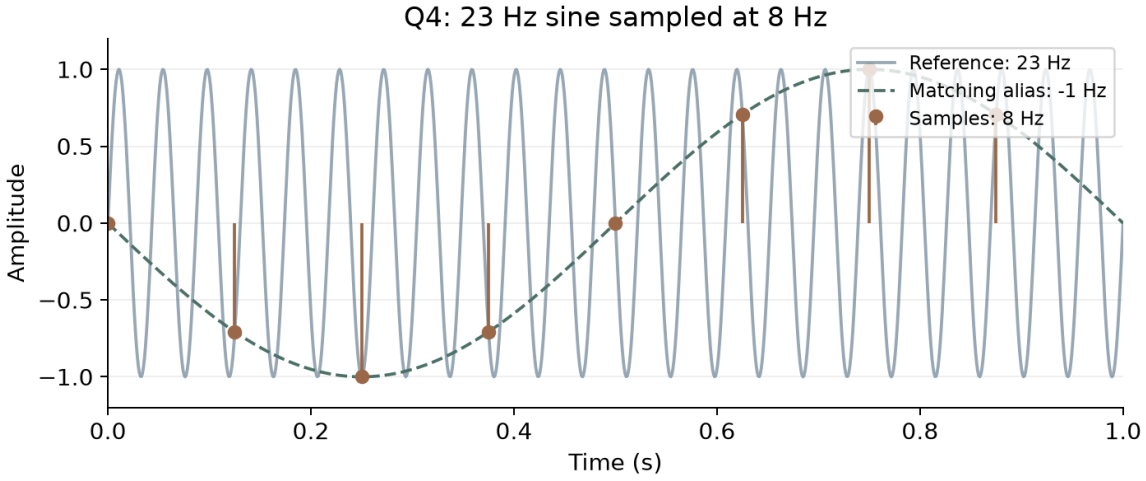


Figure 8: A 23 Hz signal sampled at 8 Hz produces a slow, reversed 1 Hz alias.

Q5. Describe a real-world scenario where understanding aliasing is crucial.

Answer: A camera can make spinning fan blades look slower or move backwards when the frame rate is too low. Choosing a high enough frame rate helps show the motion more accurately.

3.4 Image Sampling and Quantization

Q1. How does the resampling factor affect the image quality? At what factor does the distortion become apparent?

Answer: I tested resampling factors 2, 4 and 8. Factor 2 kept most details, while factor 4 made the image clearly softer and more blocky. At factor 8, the loss of detail was strong, so the distortion was easy to see.

Q2. What is the characteristic of the distortion introduced by resampling? How does it differ from the distortion introduced by requantization?

Answer: Resampling removes spatial detail and can make edges look blocky or jagged. Requantization keeps the same image size but reduces the number of gray levels, so smooth areas show banding or contour steps. The two distortions are different: one removes detail, and the other removes tone levels.

Q3. How few bits per pixel can be used in requantization before the image distortion becomes apparent?

Answer: I compared 6, 4 and 2 bits per pixel. At 6 bpp, the image looked close to the original. Banding became visible at 4 bpp and was strong at 2 bpp.



Figure 9: Grayscale Lenna: original, simple resampling (factor 4), and requantization (4 bpp).

Q4. Implement a function that applies both resampling and requantization to the image. Experiment with different combinations of factors and bits per pixel. What combination gives the best balance between compression and visual quality?

Answer: I tested four combinations. I preferred factor 2 with 4 bpp because the image still kept its main details. If the reduced pixels were stored at 4 bits each, they would need one-eighth of the original data. The images were enlarged again for comparison, so they are not compressed files.

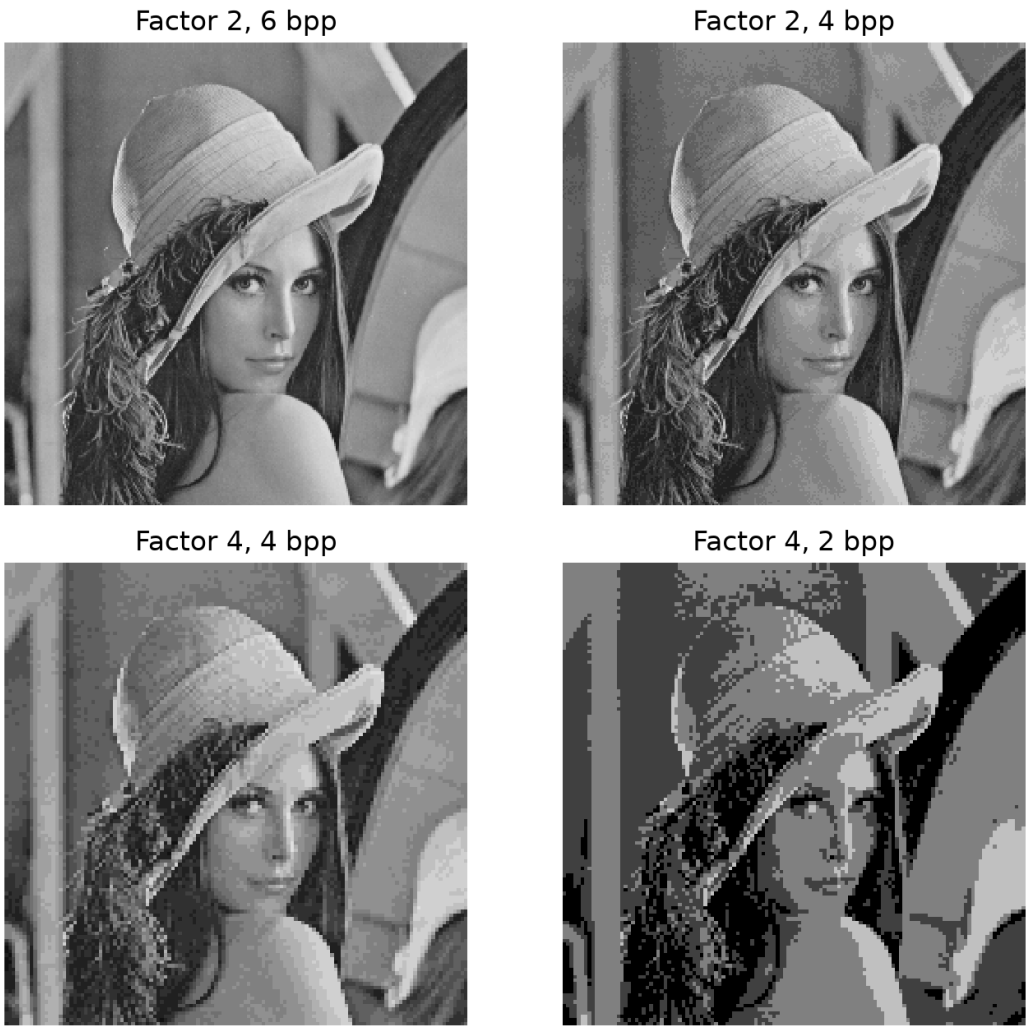


Figure 10: Combined resampling and requantization of Lenna at the four requested settings.

Q5. Modify the code to implement a more advanced resampling method, such as bicubic interpolation. Compare the results with the simple resampling method used initially. Discuss the trade-offs between image quality and computational complexity.

Answer: I compared simple and bicubic resampling at factor 4. Bicubic gave smoother edges and fewer visible blocks. It uses more calculations than the simple method, so it is generally slower.

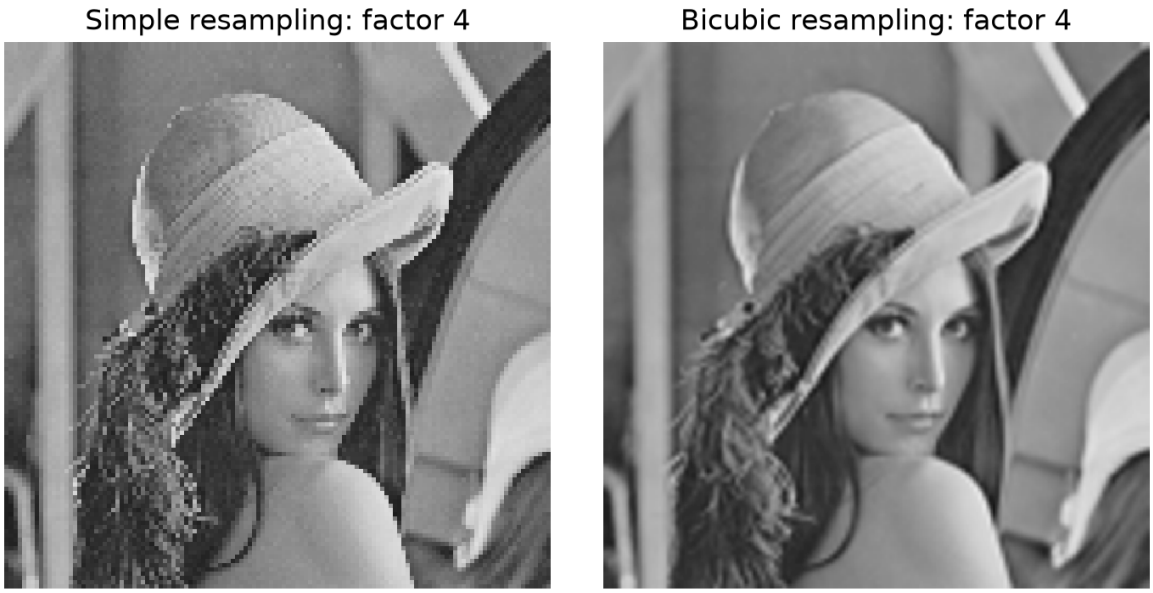


Figure 11: Simple and bicubic resampling of Lenna at factor 4.

Q6. Try using different images by changing the image_url. You can find a variety of test images at the USC-SIPI Image Database (`http://sipi.usc.edu/database/`). How do the effects of resampling and requantization vary for different types of images (e.g., portraits, textures, aerial photos)?

Answer: The portrait image kept its main shapes for longer, especially under requantization. The texture image lost fine detail sooner when I resampled it. Smooth regions showed banding more clearly, while detailed regions mainly lost texture.

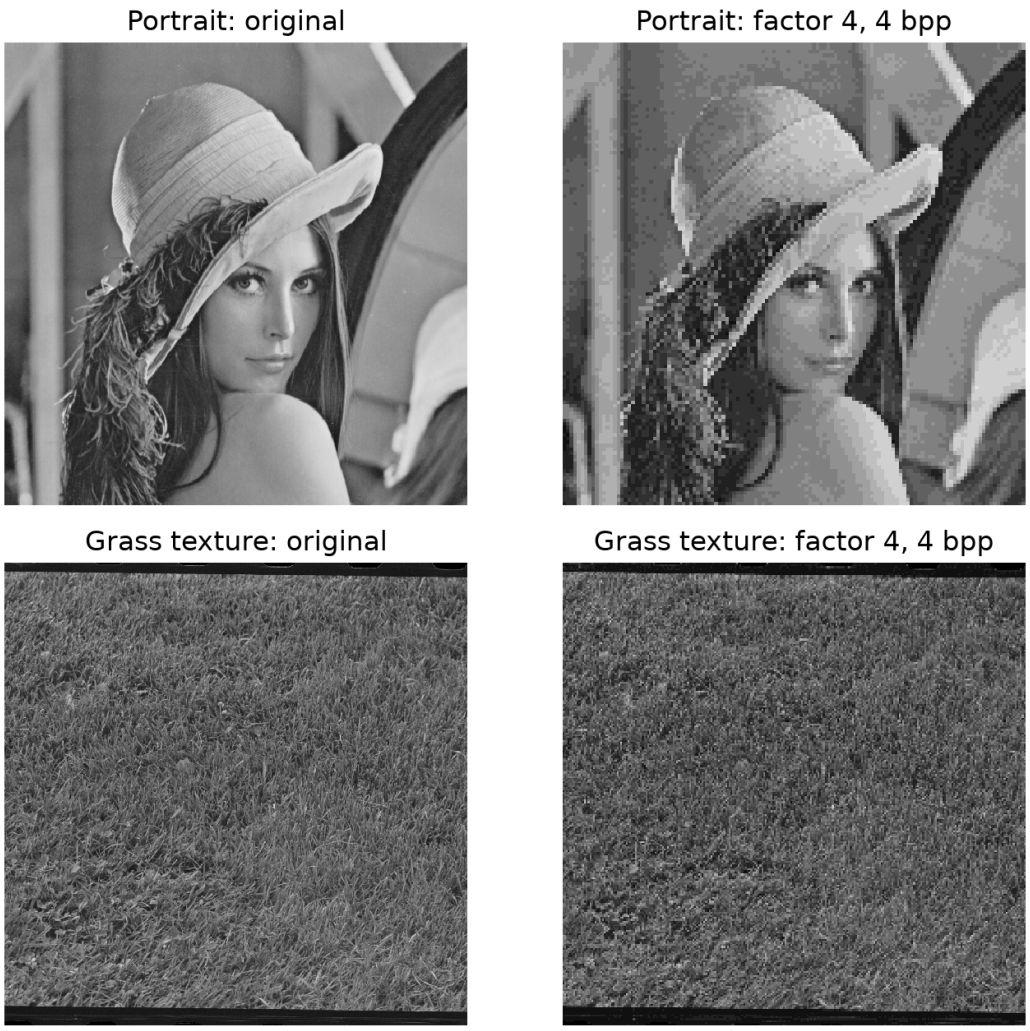


Figure 12: Portrait and grass texture before and after factor 4 resampling and 4 bpp requantization.

3.5 PSNR Calculation

Q1. Which process (resampling or requantization) introduces more distortion according to the PSNR values?

Answer: I compared the images using PSNR. Simple resampling at factor 4 had a lower PSNR than requantization at 4 bpp. This means resampling caused more pixel error in this test (Table 1).

Q2. How well do the PSNR values correlate with your visual perception of image quality?

Answer: The PSNR results mostly matched what I saw in the images. Resampling made the edges blocky, while requantization mainly caused banding. PSNR measures pixel error, but it cannot fully describe how an image looks to a person.

Q3. Create a function that calculates PSNR for different resampling factors and bits per pixel. Plot these results and discuss the trade-offs between compression and quality.

Answer: I tested different resampling factors and bit depths. Increasing the resampling factor generally lowered PSNR because more detail was lost. Using more bits per pixel improved PSNR because more gray levels were kept (Figure 13).

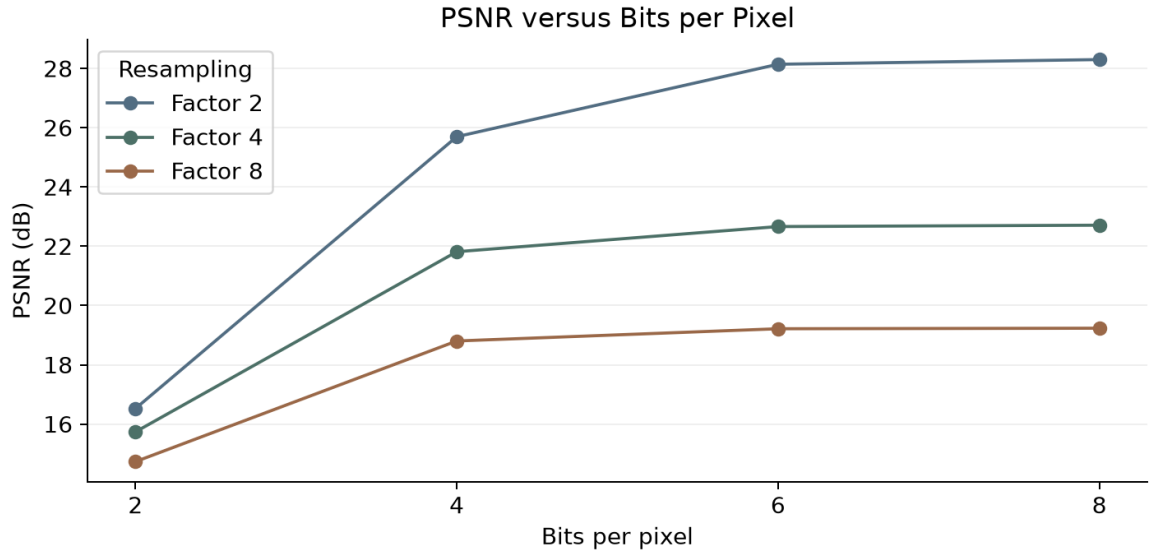


Figure 13: PSNR versus bits per pixel for resampling factors 2, 4 and 8.

Q4. Research the Structural Similarity Index (SSIM) as an alternative to PSNR. How does SSIM address some of the limitations of PSNR?

Answer: PSNR measures the average error between pixel values. SSIM also compares local brightness, contrast and structure. This helps describe image changes that PSNR may not show clearly.

Reference: Wang et al. (2004), *Image Quality Assessment: From Error Visibility to Structural Similarity.* DOI: 10.1109/TIP.2003.819861.

Q5. Implement the SSIM metric and compare its results with PSNR for your processed images. Discuss the differences in how these metrics evaluate image quality.

Answer: I calculated PSNR and SSIM for the same processed images (Table 1). A higher PSNR means less pixel error, while a higher SSIM means the image is more similar in local structure. The

two scores measure image quality in different ways.

Table 1: PSNR and SSIM for the processed Lenna images.

Processing	PSNR (dB)	SSIM
Simple resampling, factor 4	22.7098	0.6589
Requantization, 4 bpp	29.2074	0.8832
Combined factor 2, 4 bpp	25.6964	0.7524
Bicubic resampling, factor 4	28.8441	0.8282

3.6 Noise Addition and PSNR Analysis

Q1. At what PSNR value does the noise start to become visibly apparent in the image?

Answer: I added different amounts of Gaussian noise and measured the PSNR. Slight noise was visible at $\sigma = 5$, with a PSNR of 34.0821 dB, especially in smooth parts of the image (Figures 14–15).

Q2. How does this PSNR threshold compare to the often-cited 30–50 dB range for acceptable image quality?

Answer: The 30–50 dB range is only a general guide. Noise can still be visible within this range. The image content also affects how easily we notice it.

Q3. At what noise level (standard deviation) does the image become unrecognizable? What is the corresponding PSNR?

Answer: I increased the noise until the image became difficult to recognize. At $\sigma = 500$, the PSNR was 6.2490 dB. Most details were lost, although some shapes could still be seen (Figure 15).

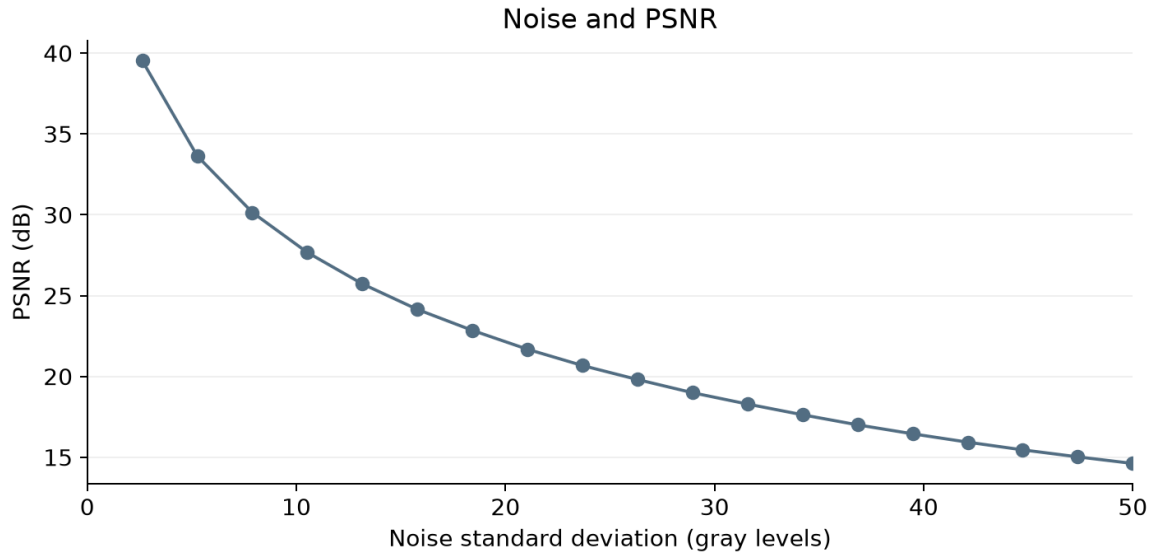


Figure 14: PSNR for 20 noise levels from 0 to 50. At $\sigma = 0$, PSNR is infinite, so that point is omitted from the curve.

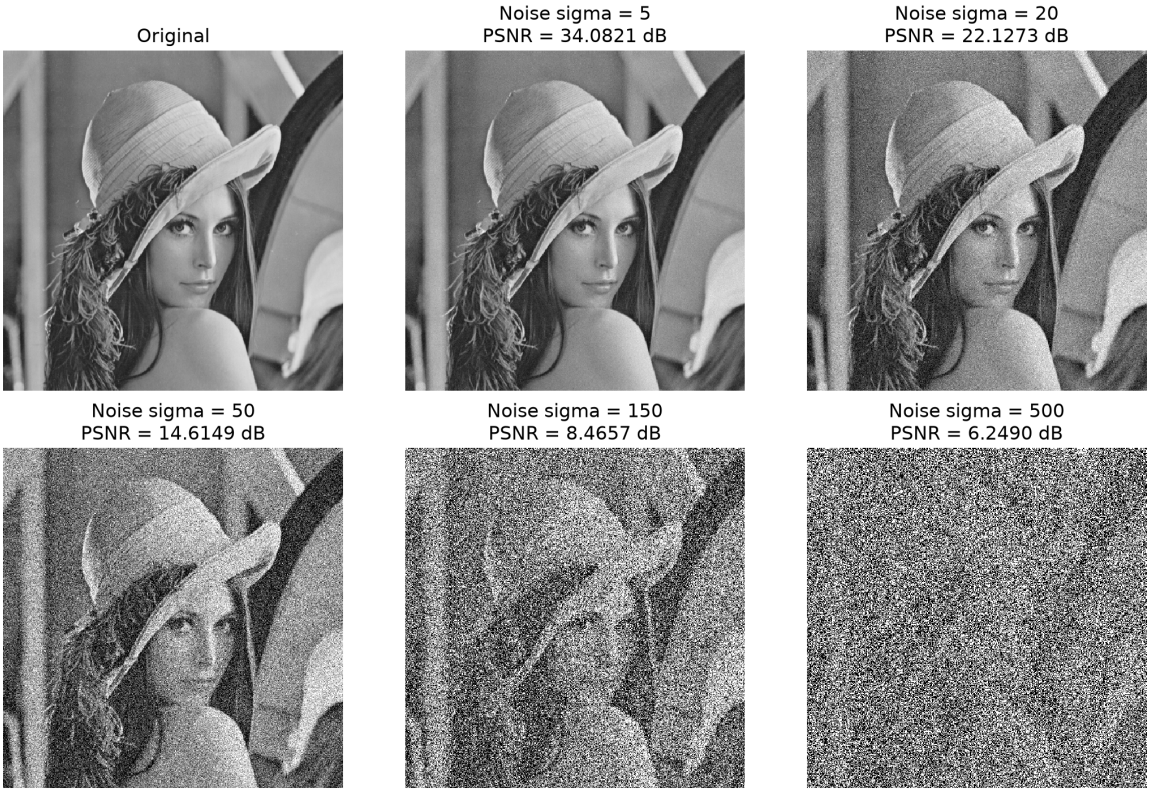


Figure 15: Original Lenna and Gaussian noise at standard deviations 5, 20, 50, 150 and 500. Each noisy image's title shows its measured PSNR.

Q4. Implement a simple noise reduction technique, such as a median filter or Gaussian blur. Apply this to the noisy images and compare the PSNR values before and after noise reduction. Discuss the effectiveness and limitations of this simple denoising approach.

Answer: I used a Gaussian filter on the image with noise $\sigma = 25$. The PSNR changed from 20.2170 dB to 29.0338 dB. The filter reduced noise, but it also made some edges and small details softer (Figure 16).

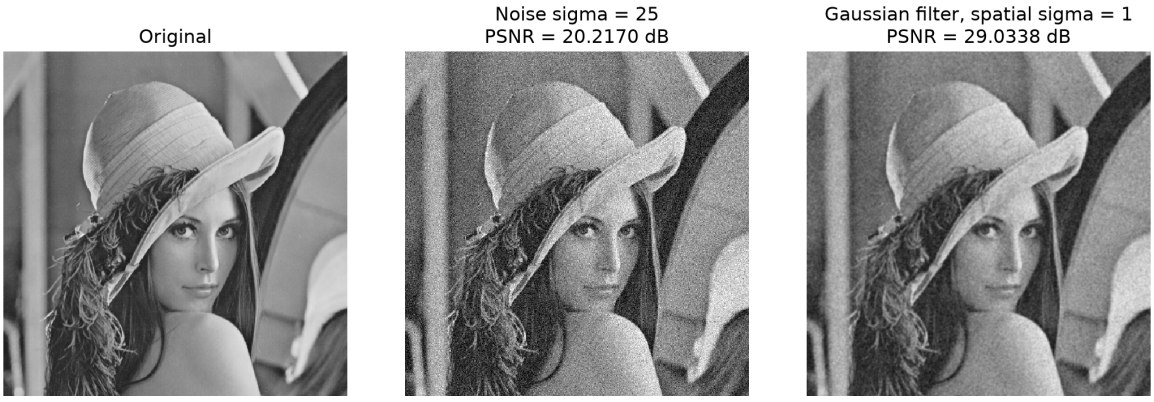


Figure 16: Original, noisy ($\sigma = 25$), and Gaussian filtered (spatial $\sigma = 1$) images. PSNR is measured against the original.

Q5. Research and briefly describe an advanced noise reduction technique. How does it improve upon simpler methods?

Answer: Non-Local Means compares small image patches and averages similar ones. It can reduce noise while keeping more edges and texture than simple blurring. It needs more calculations, so it is usually slower.

Reference: scikit-image documentation, `denoise_nl_means`: `https://scikit-image.org/docs/stable/api/skimage.restoration.html`.

3.7 Audio Signal Analysis

Q1. What is the duration of the audio clip? What is its highest frequency component?

Answer: I analyzed the piano recording. It lasted 0.5018 seconds, and its strongest FFT peak was at 219.2146 Hz. The highest frequency above my 1% threshold was 4832.6856 Hz. This is an estimate because weaker frequencies may still be present (Figure 17).

Q2. Describe the main features you observe in the spectrogram. What do the bright areas represent?

Answer: The piano sound has clear frequency bands that become weaker over time. The brighter areas show where the sound has more energy. This makes it easier to see how the sound changes (Figure 17).

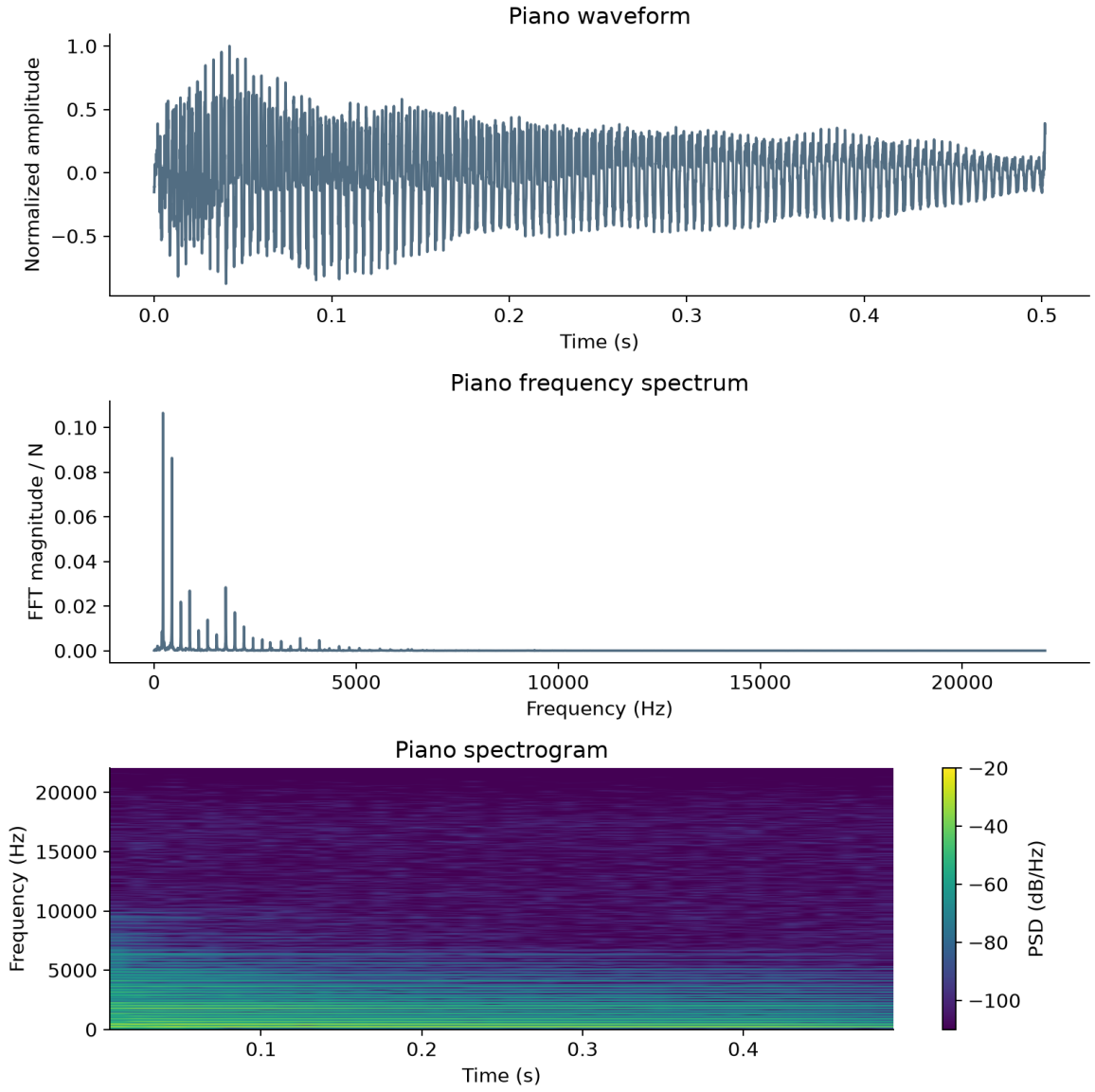


Figure 17: Piano waveform, normalized FFT magnitude and spectrogram.

Q3. Modify the code to analyze only the first half second of the audio. How does this change the frequency spectrum?

Answer: I analyzed only the first 0.5 seconds. The FFT used fewer samples, which changed its frequency spacing. The main frequency peaks stayed near the same values, but their sizes could change (Figure 18).

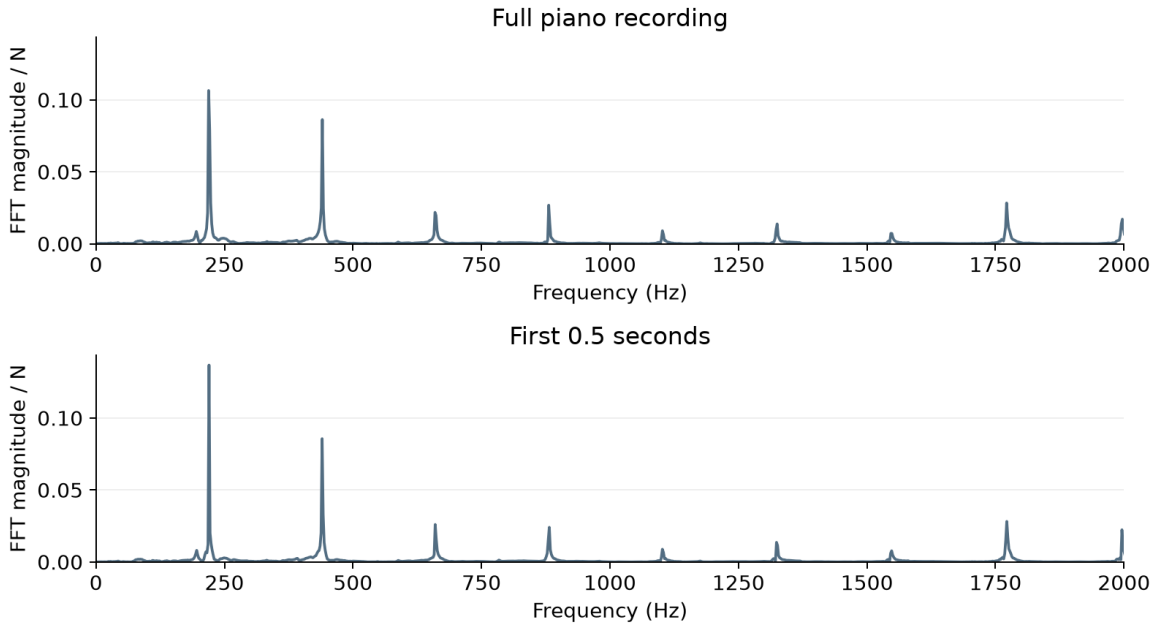


Figure 18: Full piano FFT and the first 0.5 seconds, shown over 0–2000 Hz with the same scales.

Q4. Try using different audio files by changing the audio_url. You can find a variety of sound effects and audio samples at `https://www.pacdv.com/sounds/`. How do the waveform, frequency spectrum, and spectrogram differ for different types of audio (e.g., music, speech, environmental sounds)?

Answer: I compared piano music with a thunderstorm recording. The piano had clearer repeating patterns and frequency peaks. The thunderstorm was more irregular, with sound spread over more frequencies. The spectrograms showed these differences over time (Figure 19).

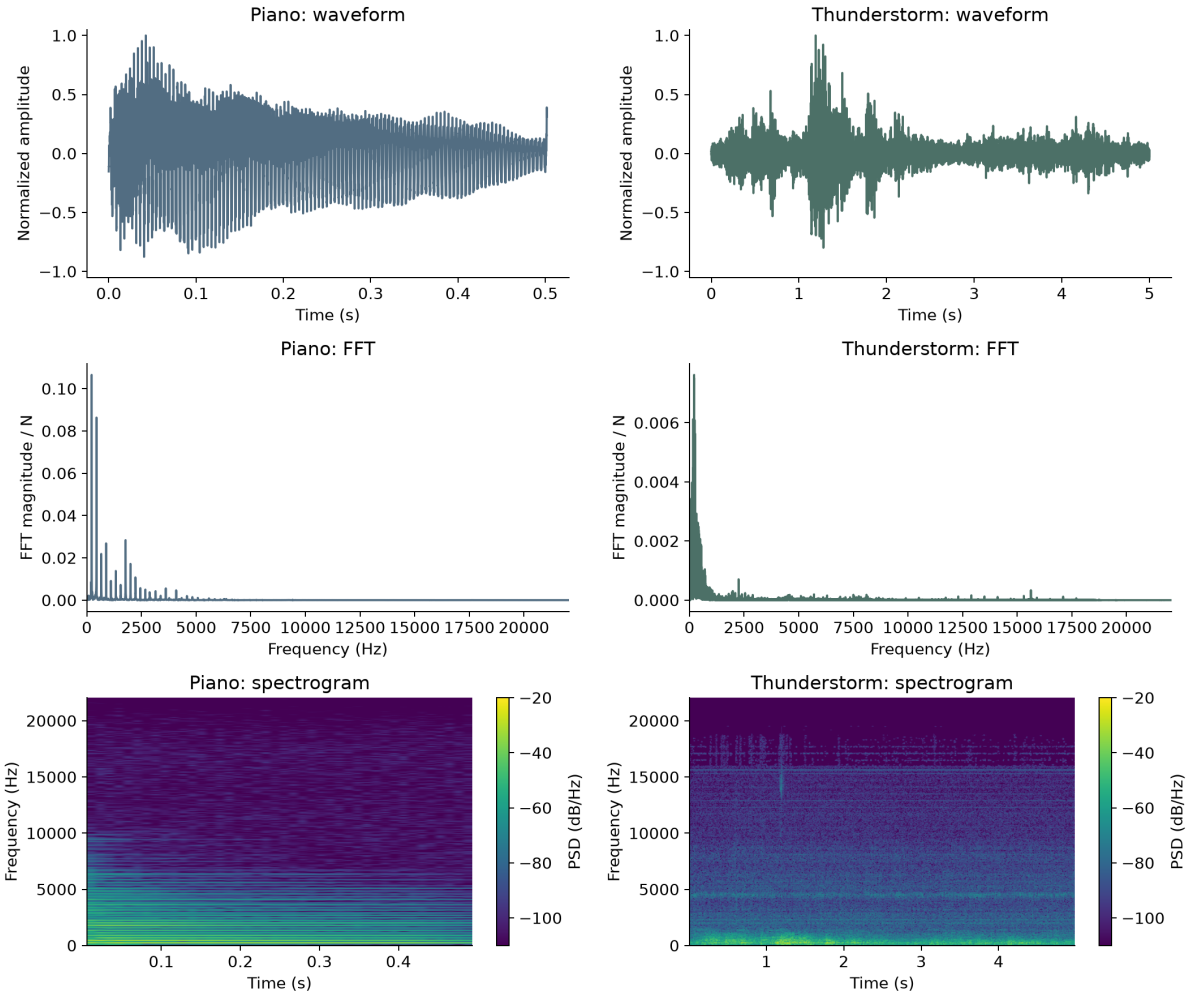


Figure 19: Piano and thunderstorm waveforms, FFTs and spectrograms. Each recording is normalized to its own peak; spectrograms use the same colour scale.

Q5. Implement a function to downsample the audio (reduce the sampling rate). How does this affect the audio quality and the frequency spectrum? At what point does aliasing become noticeable?

Answer: I reduced the audio sampling rate by keeping fewer samples. This lowered the highest frequency that could be represented correctly. In my 1,800 Hz test, sampling at 3,000 Hz produced a 1,200 Hz alias, while sampling at 2,000 Hz produced a 200 Hz alias (Figure 20). This can change how audio sounds.

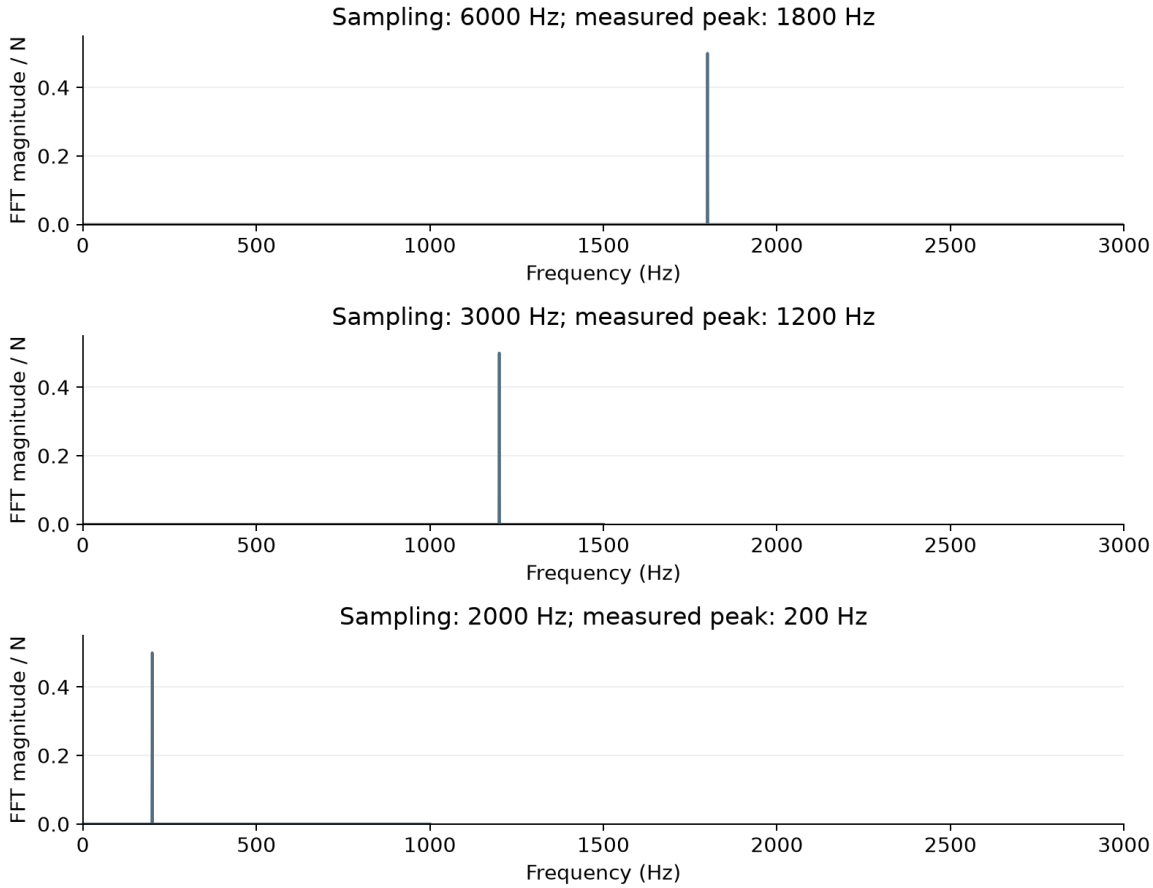


Figure 20: The 1,800 Hz test tone sampled at 6,000, 3,000 and 2,000 Hz gives FFT peaks at 1,800, 1,200 and 200 Hz.

Q6. Research and briefly describe a common audio processing technique (e.g., noise reduction, pitch shifting, time stretching). How might this technique be implemented using the concepts you've learned in this lab?

Answer: Pitch shifting changes how high or low a sound is. Playing audio samples at a different rate can change the pitch, but it also changes the duration. More advanced methods can change pitch while keeping the same length.

Reference: librosa effects documentation, `https://librosa.org/doc/0.11.0/effects.html`.

Audio sources: teacher's piano `a1.wav` and ESC-50 thunderstorm `1-101296-A-19.wav`.

4 Discussion

Changing frequency changes the number of cycles, while adding signals changes their shape. The square-wave comparison also shows why values at zero need to be checked. With added noise, the main FFT peaks stayed visible above the smaller values. When the sampling rate was too low, higher frequencies appeared as lower ones. Resampling changed image detail, while requantization changed the gray levels. PSNR and SSIM described image changes in different ways. Gaussian filtering increased PSNR for the noisy image. The tone test showed that reducing the sampling rate can produce alias frequencies.

5 Conclusion

The requested signals were generated and compared. The 20 Hz sine had twice as many cycles, and the composite had both curves and jumps. Fourier analysis identified the added 30 Hz component and the chirp's wider frequency range. The sampling experiments showed how aliasing can give a false impression of the original signal. The image comparisons showed different losses of detail and tone levels. Requantization at 4 bpp had less pixel error than simple resampling at factor 4. The noise tests showed how added noise affects PSNR. The 1,800 Hz tone appeared at lower frequencies when sampling was too slow.

Appendix: Code Snippets

```
sawtooth_wave = 2 * (5 * t - np.floor(5 * t)) - 1
square_wave_ceil = np.ceil(np.sin(2 * np.pi * 5 * t)) * 2 - 1
```

The first line removes whole cycles to make a repeating ramp. The second line is the teacher's replacement expression checked in Q3. The full script is `code/section1_1.py`. The Fourier analysis script is `code/section1_2.py`. The sampling script is `code/section1_3.py`. The image processing script is `code/section2_1.py`. The image quality script is `code/section2_2.py`. The noise analysis script is `code/section2_3.py`. The audio analysis script is `code/section3_1.py`.

Sources: DT081A Lab 1 assignment, Sections 1.1–3.1 (pages 1–5), and the DT081A lab report template.